SESE3133: Formal Methods in Software Engineering

Spring 2026

Assignment 2

Total Marks: 50 **Obtained Marks: _____** **Date: _____** **Section: F-3**

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CLO #	Keywords	Taxonomy Level
CLO 2	Use, Illustrate, Demonstrate, Change, Modify, Construct	C3

Questions Mapped	Total Marks	Marks Obtained
1	20	
2	10	
3	20	

Submission Deadline:

Hard Copy & Soft Copy: Monday, April 13, 2026 till 1:00 PM

Instructions:

- **This assignment can be done in group.**
- **No late submission accepted.**
- **Viva can be conducted from all members of the group**
- **Attempt all questions in sequence. Attach this title page as a front page of assignment.**
- **Assignment should be handwritten on A4 sized page white paper.**
- **No pages from register please.**
- **Submit Hard copy in class/office and scanned copy on the portal.**
- **This title page should be attached.**

Question no. 01

Task - 01

a) $P \vee \sim(P \wedge Q)$ is a tautology.

A tautology is a formula that is TRUE for all possible truth value assignments.

P	Q	$P \wedge Q$	$\neg(P \wedge Q)$	$P \vee \neg(P \wedge Q)$
T	T	T	F	T
T	F	F	T	T
F	T	F	T	T
F	F	F	T	T

Since, the final column is TRUE for all four rows, the formula is tautology.

b) $(P \wedge Q) \wedge \sim(P \vee Q)$ is a contradiction

Contradiction is a formula that is FALSE for all possible truth value assignments.

P	Q	$P \wedge Q$	$P \vee Q$	$\neg(P \vee Q)$	$(P \wedge Q) \wedge \neg(P \vee Q)$
T	T	T	T	F	F
T	F	F	T	F	F
F	T	F	T	F	F
F	F	F	F	T	F

Task - 02

Eaton Company - Logical Equivalence

Let G = "goods are satisfactory"
M = "money will be refunded"

Original Statement: "If goods are unsatisfactory, then money will be refunded"

$\neg G \rightarrow M$

Claim: "Goods satisfactory or money refunded"

$G \vee M$

G	M	¬G	¬G→M	G∨M
T	T	F	T	T
T	F	F	T	T
F	T	T	T	T
F	F	T	F	F

The column for "¬G→M" and "G∨M" are identical in every row. Therefore two statements are logically equivalent.

Question no. 02

TASK - 01

System Requirement in first - order Logic

Let domain = all people/things depending on context

(1)

Let C(x) : complete
V(x) : verified
I(x) : implemented

$\Rightarrow (C(x) \wedge V(x)) \rightarrow I(x)$

(2)

Let A(x) : can access
P(x) : premium

$\Rightarrow A(x) \rightarrow P(x)$

(3)

Let S(x) : Student
FM(x) : takes FM
A(x) : takes Automata

$\Rightarrow \forall x [(S(x) \wedge FM(x)) \rightarrow A(x)]$



(3)

Let OnTime(x)

$\Rightarrow \exists x \; OnTime(x)$

(5)

Let $R(x)$ = Registered

$RN(x)$ = has roll number

$\Rightarrow \forall x \; [R(x) \rightarrow RN(x)]$

(6)

let $W(x)$ = working

$F(x)$ = feature

$\Rightarrow \exists x \; [F(x) \wedge \neg W(x)]$

(5)

TASK - 02

Role of Quantifiers

1) Universal Quantifiers (∀)

Means "for all"

Example:

∀ x Student(x) → Studies(x)

=> Every student studies.

2) Existential Quantifiers (∃)

Means "there exists"

Example:

∃ x Student(x) ∧ Smart(x)

⇒ atleast one student is smart.

Question no. 03

Temporal Logic

Task-01: Modify following statements in Temporal syntax

1- R1 never holds in two consecutive states
⇒ If R is true now - next state must be false

- Formula: $G(R \rightarrow X \neg R)$

2- R holds in at most one state
⇒ means: once R happens → it never happens again

Formula: $G(R \rightarrow G \neg R)$

4- If R holds in a state, it holds forever after
⇒ If R now → S was true in (i-1) or (i-2)

formula: $G(R \rightarrow (\text{previous } S \vee \text{previous previous } S))$
or $G(R \rightarrow (S \text{ at } i-1 \vee S \text{ at } i-2))$

3- If R holds at si_S must hold in one of previous 2 states
~~⇒ If R now S was true in (i-~~
⇒ If R becomes true → always true

Formula: $G(R \rightarrow GR)$

5- System always prevents unauthorized access.
⇒ bad thing NEVER happens
formula $G \neg U$

6- System eventually completes processing
⇒ Mean :- at some future time → completed

formula: $F C$

Task-2 : LTL / CTL

1 - α appears eventally on all paths

all paths + eventually

Formula : $AF\ \alpha$

2 - α holds untill δ appears on all paths

α continues unit δ

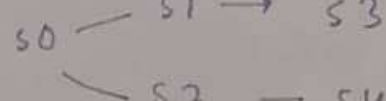
Formula : $A\ (\alpha\ U\ \delta)$

3 - $EF\ p$

=> Already given — means :

=> There exists a path where eventually p happens

Graph :



4 - X_p

In the Next state, p must be true

Graph :

